

OTTO LILIENTHAL

BORN IN POMERANIA (part of present-day Poland) in 1848, Otto Lilienthal was fascinated from an early age by bird flight. Although he trained as an engineer and ran a factory building steam engines, he remained convinced that ornithology held the key to human flight – a belief reflected in the title of his 1889 publication, *Birdflight as the Basis of Aviation*. Although he became famous for his experiments with what would now be called hang gliders, flying his first one

in 1891, he never abandoned the idea of flapping wings as a means of propulsion. In his systematic work on wing shapes, Lilienthal showed a genuinely scientific temperament, but he also possessed a streak of showmanship that helped publicize the pursuit of human flight. He was also a man of great physical courage who had a huge impact on the development of flight. Lilienthal died on August 10, 1896, following a glider crash.

PIONEERING BIRDMAN

Despite his uncompromising methods and “birdman” reputation, Lilienthal’s intelligent and systematic approach to flying made him a powerful inspiration to other serious researchers.



they take to the air. Contemporary experimenters in unpowered flight, by contrast, hoped to make progress through building up experience of flying. Their acknowledged leader was the German “flying man” Otto Lilienthal.

The flying man

In some respects Lilienthal was a direct descendant of the medieval “birdmen” and “tower jumpers.” Lilienthal’s flights – pacing down a hill into the wind, encumbered by his wide birdlike wings, and lifting into a glide that carried him high above the ground – were an impressive spectacle but not consonant with the notion of progress as understood in the late 19th century. It is easy to see why Hiram Maxim, with his powerful state-of-the-art steam engines and expensively constructed test track, dismissed Lilienthal as a “flying squirrel.” Yet the apparently eccentric Lilienthal was far more scientific and practical in his exploration of flight than either Maxim or Ader. From a scrupulous study of bird flight and bird anatomy, he concluded that a curved, or cambered, wing was essential to produce lift. He

IMPRESSIVE SPECTACLE

Otto Lilienthal’s glider experiments attracted substantial crowds and won him a reputation as “the flying man.” Most of the tests were carried out at Lichterfelde, outside Berlin, where Lilienthal built a conical hill that allowed him to take off from any side, responding to the direction of the wind.

They were light and flimsy structures, made by stretching cotton material over willow and bamboo ribs. But, unlike Maxim’s ponderous machine, they actually flew.

In all, Lilienthal carried out more than 2,000 flights, the longest covering a distance of 1,150ft (350m). A key lesson he learned from these

experiences was that the air could be a treacherous medium to move through.

Since Lilienthal’s gliders had no control system, he was obliged to throw his body around to maintain balance and stability amid the shifting air currents.

A visiting American journalist described Lilienthal’s energetic mode of flight: “He went over my head at a terrific pace, at an elevation of about 50 feet [15m]... The apparatus tipped sideways as if a sudden gust had got under the left wing... then with a powerful throw of his legs he brought the machine once more on an even keel, and sailed away across the fields at the bottom.”

The ultimate sacrifice

Repeated flights in such unstable machines involved an astonishing level of risk. Lilienthal did devise a shock absorber to protect him if he crashed, but only used it fitfully. On August 9, 1896, caught in a sudden gust of wind, his glider stalled and crashed. He died of his injuries the following day. By then Lilienthal was a famous man. Photographs of him in flight had inspired a great deal of public interest, and his writings had been translated into several languages. His most successful glider, the No. 11 standard monoplane, had been sold to a number of enthusiasts, making it the first aircraft to be produced in quantity.

